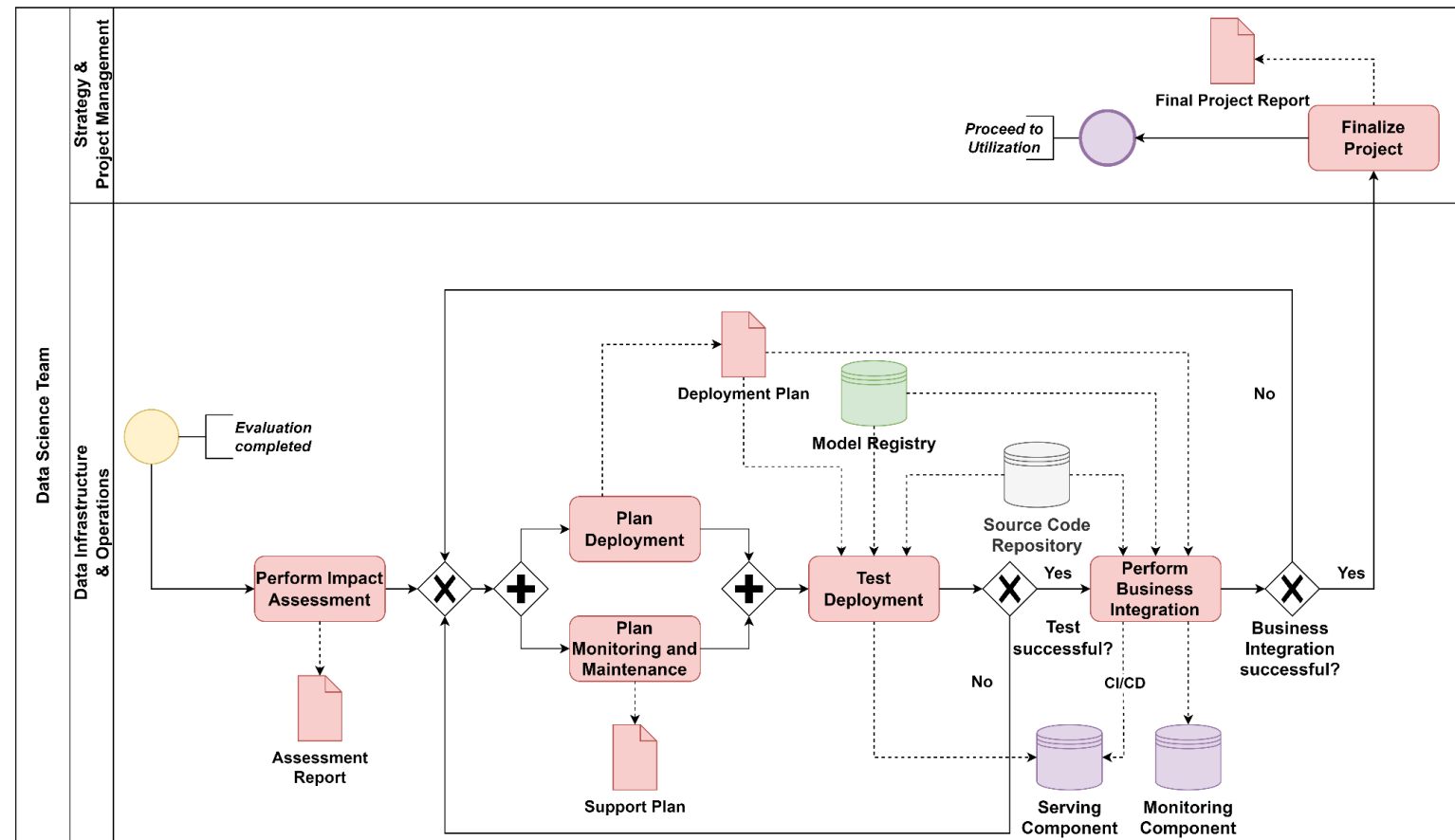


Deployment

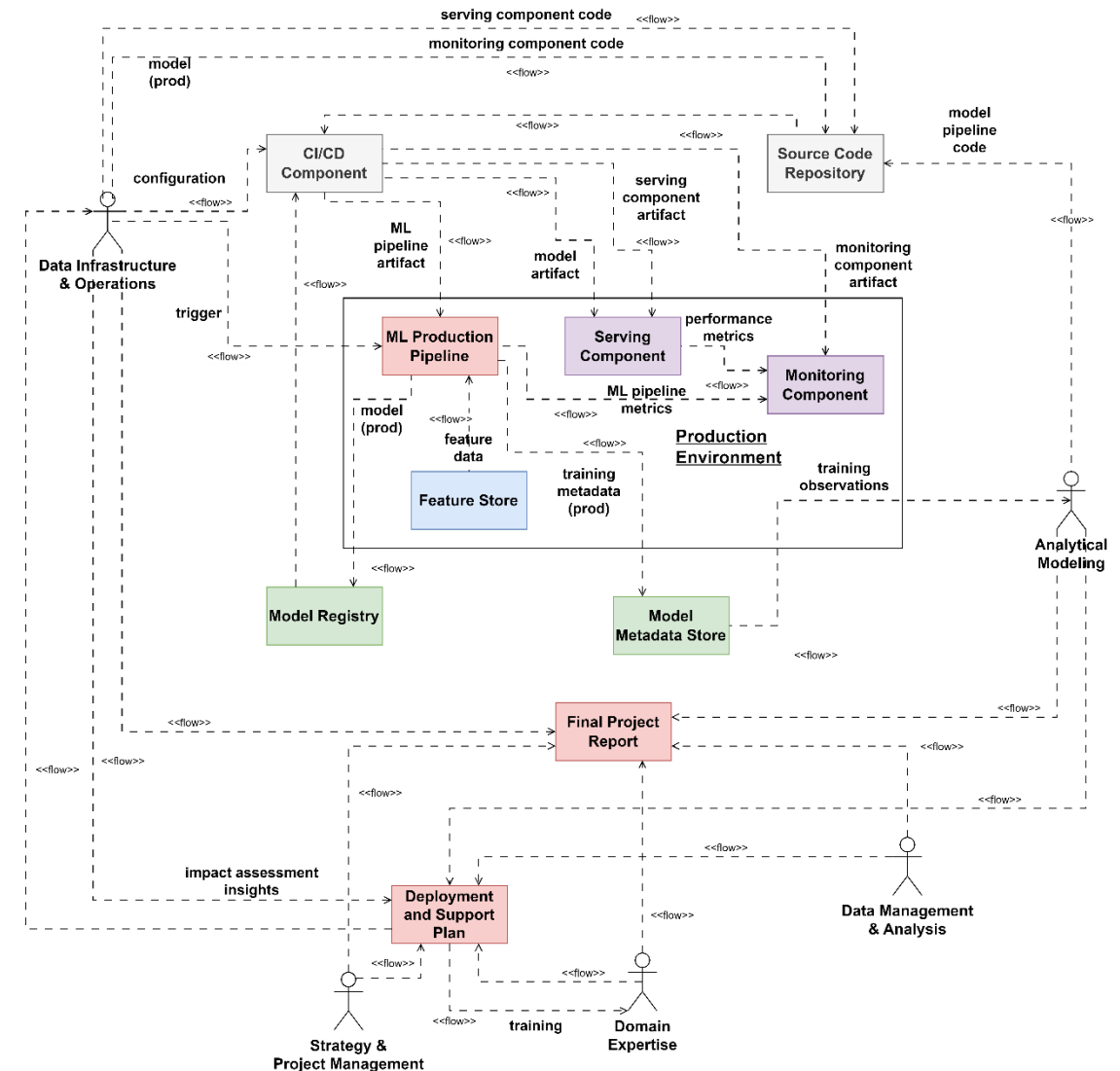
Deployment – Process Perspective

- Plan and execute the deployment of analytical artifacts
- Main tasks
 - Perform Impact Assessment
 - Plan Deployment
 - Plan Monitoring and Maintenance
 - Test Deployment
 - Perform Business Integration
 - Finalize Project
- Artifacts
 - Assessment Report
 - Deployment Plan
 - Support Plan
 - Final Project Report



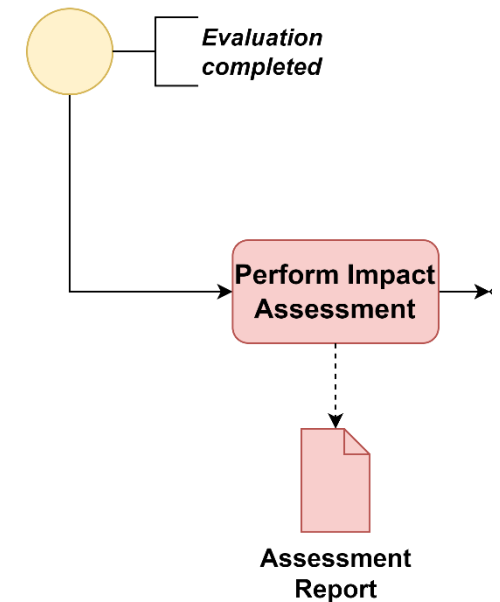
Deployment – Information Flow Perspective

- Deployment stage is significantly shaped by the defined form of the final product
- Includes the ML pipeline, serving component, and other system parts relevant for monitoring and maintenance
 - Influenced by Deployment and Support Plan
 - MLOps components support system adaptability and transparency



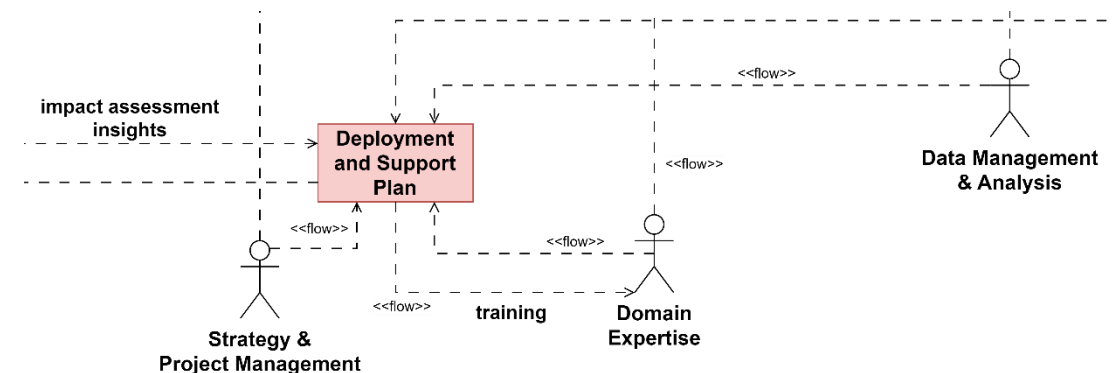
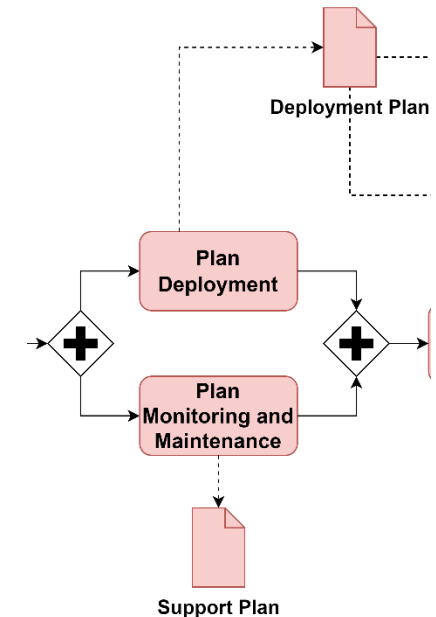
Perform Impact Assessment

- Goal: determine and refine requirements for deployment based on insights from the previous project phases
- Details the functional and operational requirements initially defined during Business Understanding
 - Needed resources and technical feasibility
- Assessment Report
 - Data
 - volume, processing requirements, handling of data sources, automation, etc.
 - Timeliness of analytics
 - Hardware and software stack
 - Technical and regulatory conditions



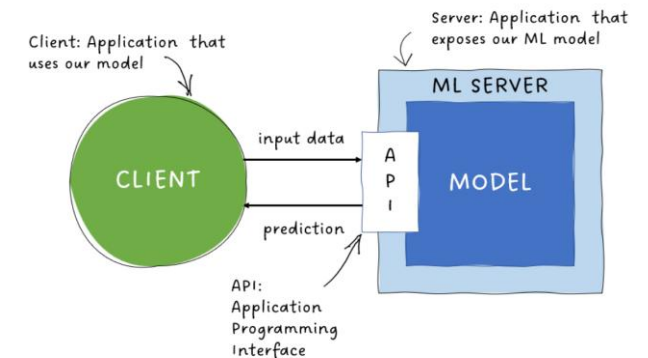
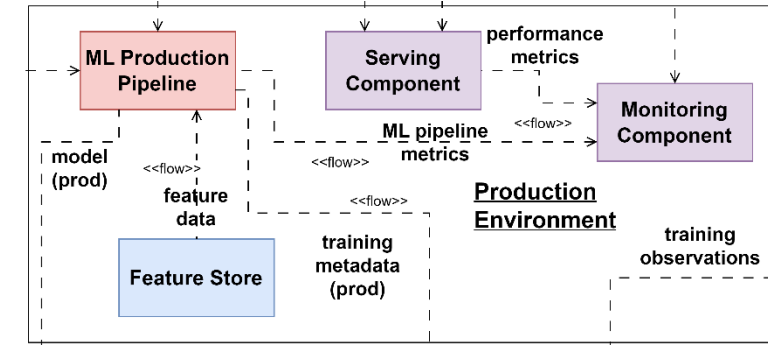
Plan Deployment

- Goal: create a roadmap of the deployment approach
- Develop and evaluate alternatives for deployment of the results
- Deployment Plan
 - Preparatory tasks
 - Regulatory considerations, process integration, etc.
 - Architecture plans
 - Pipelines, serving component, etc.
 - Test plans
 - Application, pipeline, and model tests
 - Deployment tests
 - Deployment resource and time plan
 - User enablement
 - Training plans
 - User guide



Serving Component

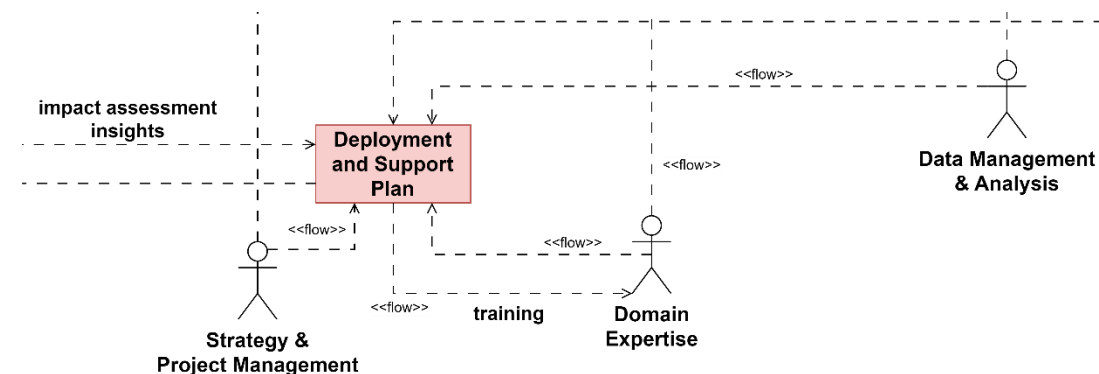
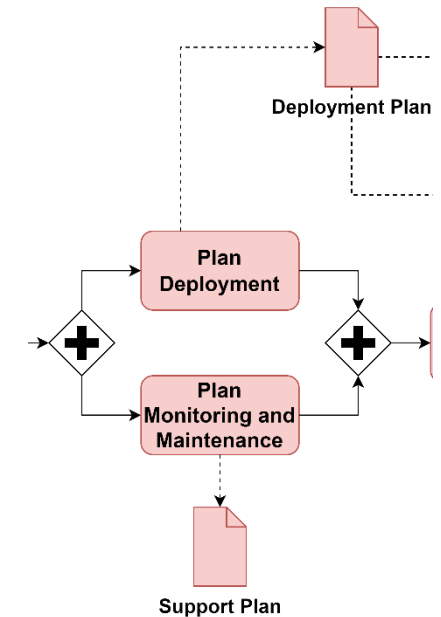
- Deployed model is available for inference via a serving component
 - Inferencing: process of sending new input to the model and receiving output from the model
- scalable and distributed model serving infrastructure recommended
- Can be configured for different purposes, examples:
 - Serving via REST API
 - Model containerization with Kubernetes and Docker
 - Web application frameworks (Flask)
 - ...
- Inference mode
 - Online: for (near) real-time predictions
 - Batch: using large volumes of input data



Model request and response through an API
(source: DataCamp)

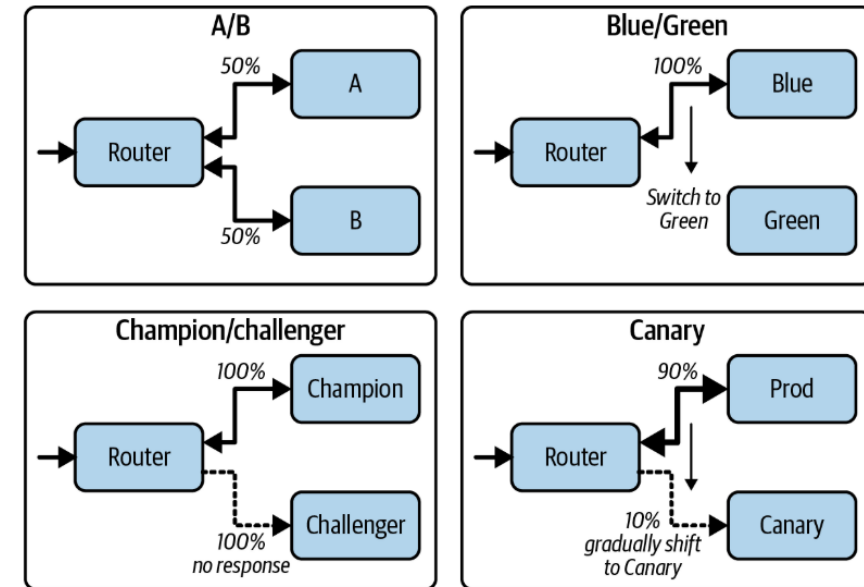
Plan Monitoring and Maintenance

- Goal: Define measures for maintenance of the DS solution
 - Involves deployed model(s), pipelines, and overall application
- Monitoring component as the central MLOps component
 - Realizes model maintenance together with the deployed ML pipeline (can take different forms)
- Support Plan
 - Parameters for evaluation of analytical artifacts and data
 - Performance thresholds
 - Countermeasures for model maintenance/retraining
 - Application metrics
 - Performance thresholds
 - Countermeasures for system maintenance
 - Usage data
 - Thresholds for appropriate use
 - Countermeasures in case of issues
 - Error handling
 - Countermeasures in case of system failures



Plan Deployment

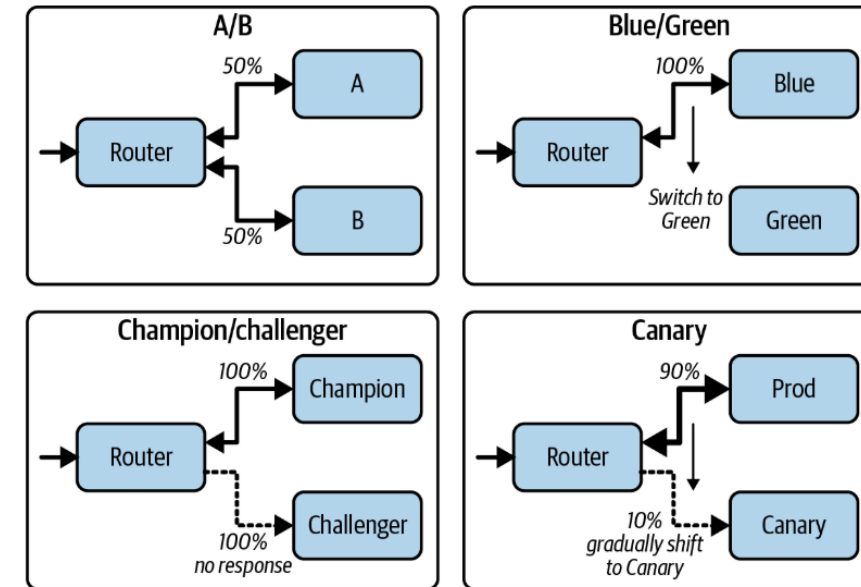
- Model deployment strategies (serving component)
 - A/B testing
 - new model deployed alongside the old model, and traffic is divided between the two
 - output of both models is monitored to determine which one performs best, and the best-performing model is promoted
 - Blue/Green
 - Setting up two identical environments (Blue and Green), with one being the live production environment and the other a staging environment
 - new model is deployed to the staging environment, once it operates as expected, staging becomes production, the old version is removed, and more recent models are deployed into staging
 - high operational costs (two identical environments)



Model deployment strategies (Haviv & Gift 2023)

Plan Deployment

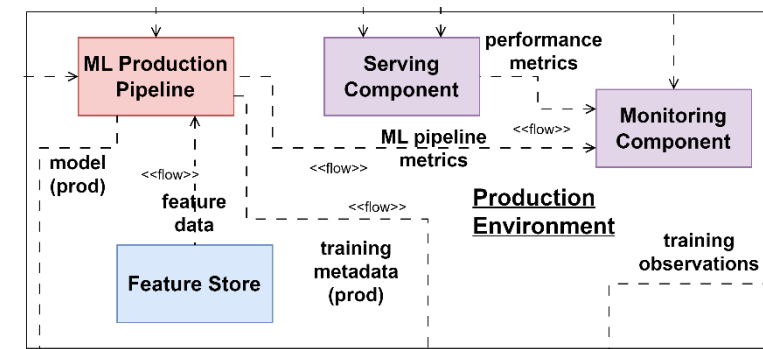
- Model deployment strategies (serving component)
 - Canary
 - Incremental rollout of new model to user subset to validate performance
 - Once validated, the model is gradually rolled out to more users
 - strategy enables controlled rollout with no downtime
 - slow, testing occurs in production → recommended for noncritical applications
 - Champion/Challenger (Shadow)
 - Deploying the champion model to production alongside one or more challenger models
 - Traffic routed to all models and the output of the champion used
 - challenger's output is only monitored
 - highest form of model validation, yet most expensive.



Model deployment strategies (Haviv & Gift 2023)

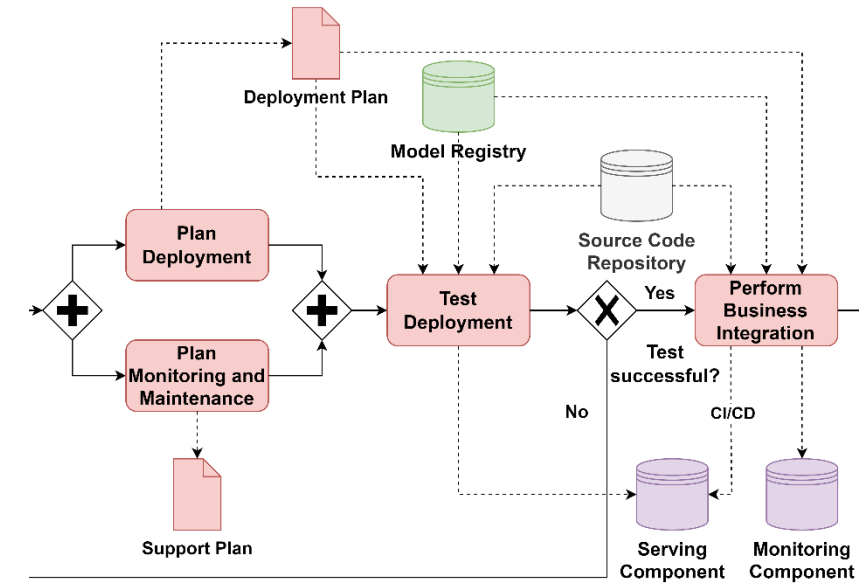
Monitoring Component

- Responsible for continuous monitoring of model serving performance, infrastructure, CI/CD, etc.
- Examples
 - Resource monitoring
 - CPUs, GPUs, memory, storage used
 - Application health and availability of the service(s)
 - Model and data monitoring
 - Model performance and data used by the model (pipeline) → accuracy, drift, bias, data quality, etc.
 - Application/pipeline monitoring
 - Overall application performance across all pipeline steps (throughput, latency, errors, etc.)
 - Business KPIs
- Tools: Prometheus with Grafana, ELK stack (Elasticsearch, Logstash, Kibana), TensorBoard, etc.



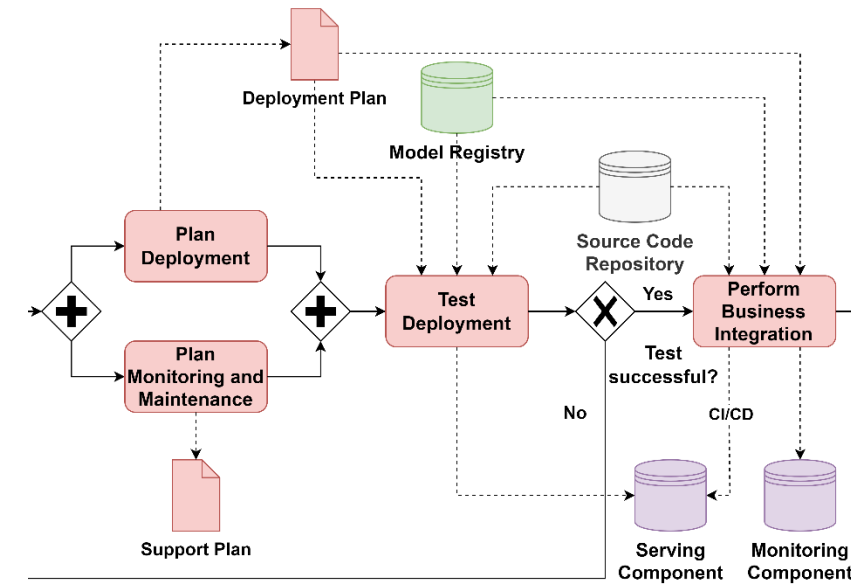
Test Deployment

- Goal: conduct tests as defined in the deployment plan before business integration
- Beyond model, pipeline, and system tests: includes test of the roll-out itself
 - Prediction latency
 - Model artifacts correctly loaded
 - Smoke tests for API endpoints
 - ...
- Ideally facilitated by CI/CD component during earlier project phases
- Documentation of test results
- Decision on proceeding with business integration depends on test results



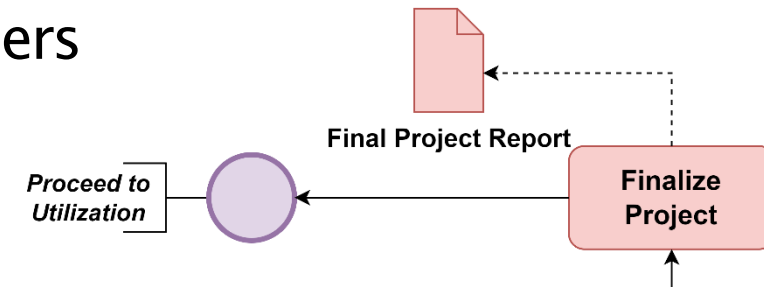
Perform Business Integration

- Goal: finalize integration of the DS solution into production
 - Includes data and model pipelines
- Model – produced by ML pipeline and versioned in model registry – is made available on the serving component → application/users can access model for predictions
 - Based on deployment strategy
- User training to ensure adequate use for effective business value generation
 - Implementation of an active solution support may be required



Finalize Project

- Goal: review and finalize project
- Official closure of the project (not the DS lifecycle)
- Assessment of best practices, pitfalls, and improvements
- Can involve a final presentation to management stakeholders
- Final Project Report
 - Executive summary (non-technical)
 - Description of application domain
 - Project management/controlling
 - Description of individual project phases
 - Conclusion
 - Results (e.g., metrics such as costs/benefits)
 - Lessons learned
 - Next steps
 - Appendix



References

- Chapman, P., Clinton, J., Kerber, R., Khabaza, T., Reinartz, T., Shearer, C., & Wirth, R. (2000). CRISP-DM 1.0: Step-by-step data mining guide. SPSS.
- Haertel, C., Pohl, M., Staegemann, D., & Turowski, K. (2022). Project Artifacts for the Data Science Lifecycle: A Comprehensive Overview. 2022 IEEE International Conference on Big Data (Big Data).
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- <https://app.datacamp.com/learn/skill-tracks/mlops-fundamentals>
- Kreuzberger, D., Kühl, N., & Hirschl, S. (2023). Machine Learning Operations (MLOps): Overview, Definition, and Architecture. IEEE Access, 11, 31866–31879.